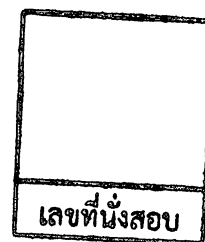


College of Industrial Technology

King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2017

Subject: 392132 Physics II

Section: 15-18

Date: 8 May 2018

Time: 10.00 - 12.00

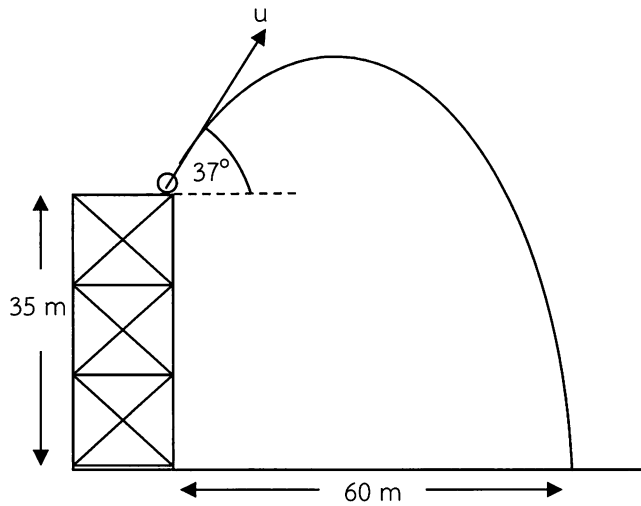
Name: \_\_\_\_\_ ID: \_\_\_\_\_ Class: \_\_\_\_\_

Instructions:

1. The examination has 10 pages (including this page) and a total score of 70 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are **NOT** allowed in the examination.
9. Electronic communication devices are **NOT** allowed in the examination room.
10. If not specified otherwise, used these values:  $g = 10 \text{ m/s}^2$ ,  $\pi = 3.14$ ,  $\pi^2 = 10$ .

**Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.**

1. A man on the tower in the figure throws a ball upward at an angle of  $37.0^\circ$  to the horizontal. The point of release is 35.0 m above the ground. The ball hits the ground 60 m from the tower. Determine velocity in m/s, is the ball thrown? (8 marks)



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2. A wheel rotates with a constant angular acceleration of  $6\pi$  rad/s<sup>2</sup>. If the angular speed of the wheel is  $2\pi$  rad/s at  $t = 0$  s,

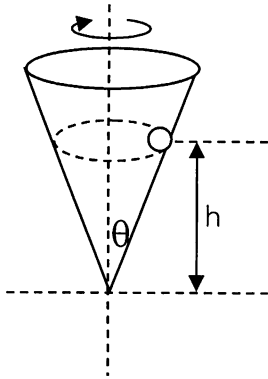
- (a) Through what angle does the wheel rotate between  $t = 0$  and  $t = 2.00$  s? Give your answer in radians and in revolutions. (4 marks)
- (b) Determine the angular speed of the wheel at  $t = 2.00$  s? (2 marks)
- (c) Determine angular displacement (in revolutions) results while the angular speed found in part (b) doubles? (4 marks)

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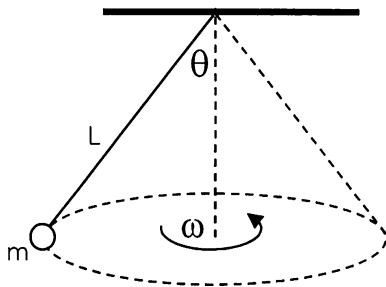
3.1 A ball mass  $m$  moving in circle path inside a cone with high  $h$  by constant speed  $v$ . find

(a) Time period (1 marks)      (b) Frequency (1 marks)

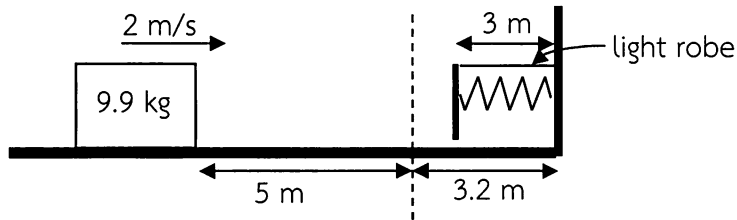
(c) Force of cone act on the ball (2 marks)



3.2 A small metal ball mass  $m$  is suspended from the ceiling by a rope of negligible mass. The ball is then set in motion in horizontal circle so that the rope describes a cone with angular speed  $\omega$  and tension force of rope is  $T$ . Show tension force proportional with angular speed square. (5 marks)



4. A spring has original length of 3.2 m and tied by light rope to the wall at length of 3 m. A box mass 9.9 kg starts moving 5 m from the spring with a speed of 2 m/s on a smooth horizontal plane to attach to the spring shown in figure. Determine is the distance of the box far from the wall before it reverses? (Spring constant is 10 N/m) (10 marks)



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5. A ball of 0.5 kg is kicked at an angle of  $53^\circ$  from the ground with an initial velocity  $U$ .
- (a) Determine the initial velocity  $U$  of the ball if its kinetic energy is 25 J at its maximum high? (5 marks)
- (b) Determine the maximum height reached by the ball? (5 marks)

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6. A 300 kg piano slides 10 m down a  $37^\circ$  incline from a rest and is kept from accelerating by a man who is pushing back on it parallel to the incline to the constant velocity 4 m/s. The effective coefficient of kinetic friction is 0.4. Calculate

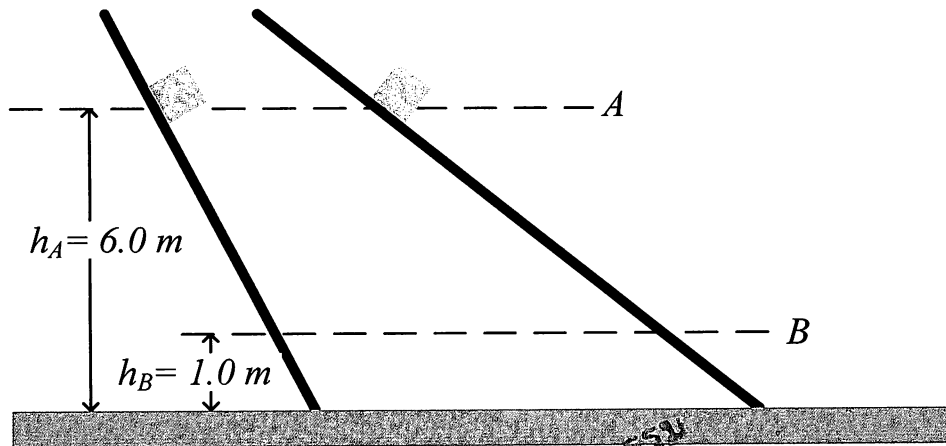
- (a) the force exerted by the man. (2 marks)
- (b) the work done by the man on the piano. (2 marks)
- (c) the work done by friction force. (2 marks)
- (d) the work done by force of gravity. (2 marks)
- (e) the power by the force exerted by the man. (2 marks)

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(7.1) Two boxes are resting on the frictionless ramps. One box is relatively light and sits on a steep ramp. The other box is heavier and rests on a ramp that is less steep. The boxes are released from rest at A and allowed to slide down the ramps. The two boxes have masses of 11 and 44 kg. If A and B are 6.0 and 1.0 m, respectively, above the ground, determine

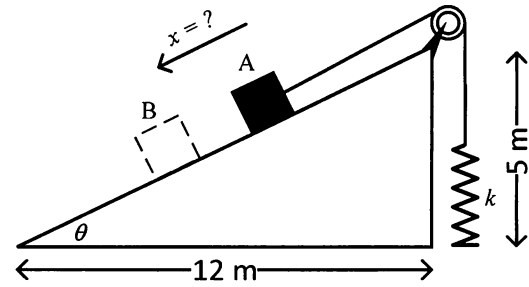
(a) The speed of the lighter box. (2 marks)

(b) Determine is the ratio of the kinetic energy of the heavier box to that of the lighter box at B. (2 marks)





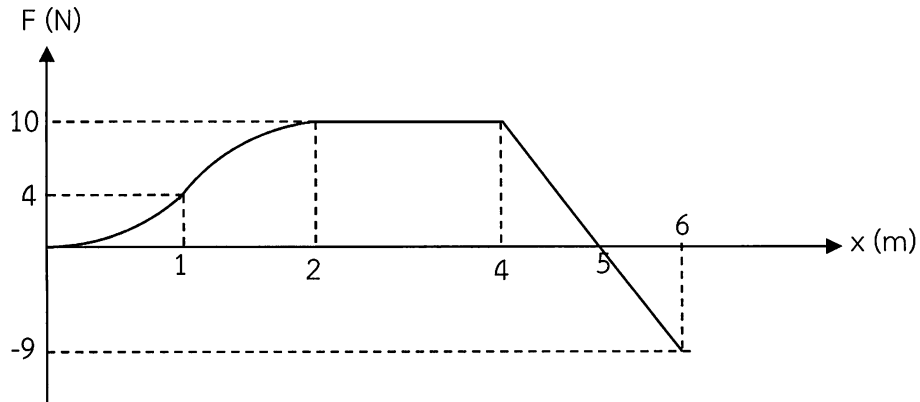
(7.2) In figure as shown, the pulley has negligible mass, and the inclined plane are frictionless. A block has a mass of 2.6 kg, which is connected with a spring. The spring has a spring constant,  $k = 400$  N/m. If the block is released from rest at A that the spring does not stretch. Find the solution from these questions:



- Determine is maximum distance  $x$  when the block reach at B? (3 marks)
- The difference of  $x$  in (a) and maximum distance  $x$  when the inclined plane is friction coefficient = 0.13. (3 marks)

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8. A man pull the box by force  $F$  with angle  $\theta$  from the ground. The force  $F$  of a man varies with distance  $x$  shown in figure. Find the work done of box between distance  $x = 2$  m to 5 m. (3 marks)



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